

Modeling slug rise at Stromboli, and the associated VLP seismic signal.

C. P. Montagna, S. Colucci

Istituto Nazionale di Geofisica e Vulcanologia, Pisa, Italy

March 22, 2022

Methods

- gas slug ascent at Stromboli modeled using MagmaFOAM
- pressure at the boundaries of the fluid system used to calculate ground displacement at the surface
- ground deformation calculated by Green's functions integration by syntheticSignals

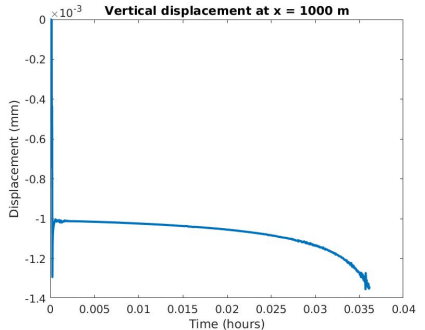
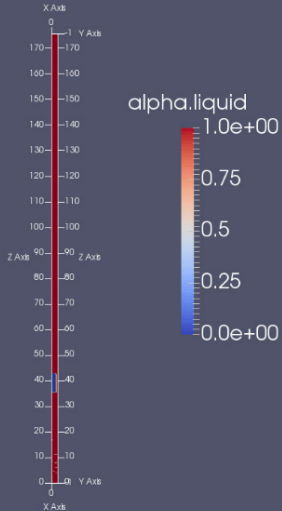
Simulations

- gas either injected continuously or placed at the bottom of a tube-like conduit
- conduit is 170 m long and 1 m wide
- magma has constant, homogeneous density and viscosity

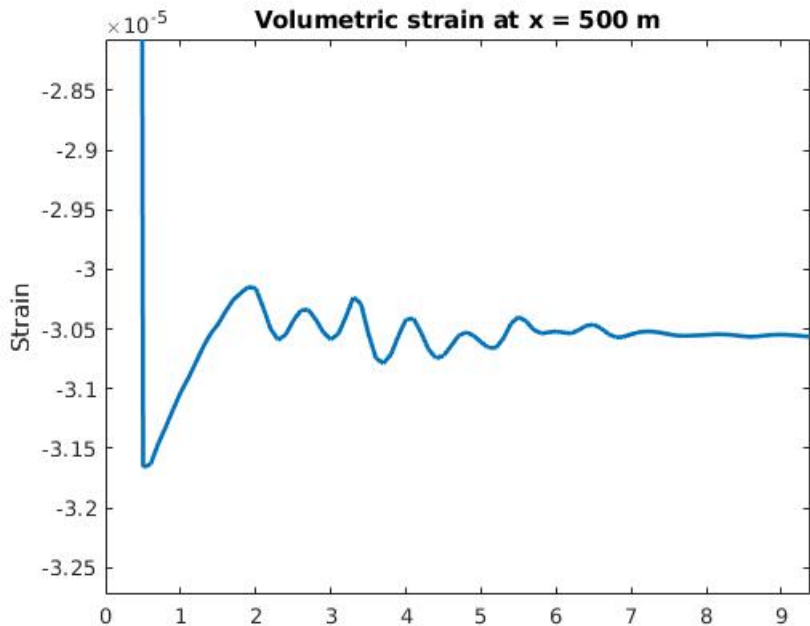
Simulation 1

batch of gas rising from the bottom

Time: 47.500000



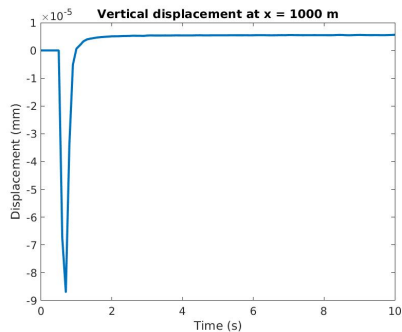
Simulation 1



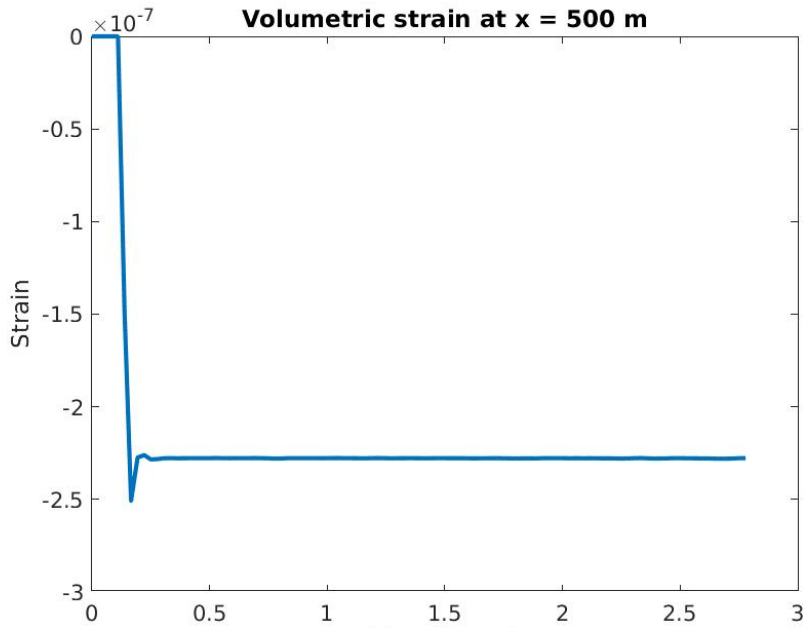
Simulation 3

gas input, different velocity (slower)

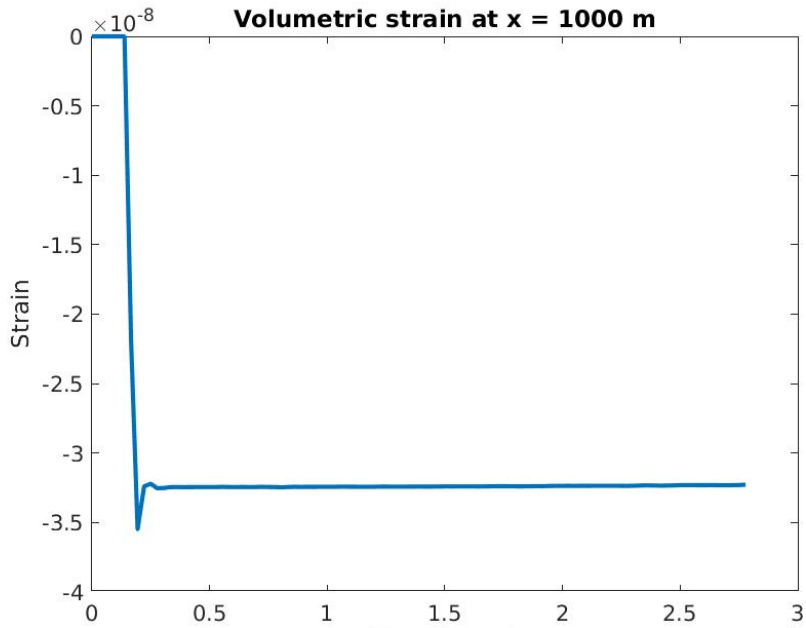
Time: 8.900000



Simulation 3

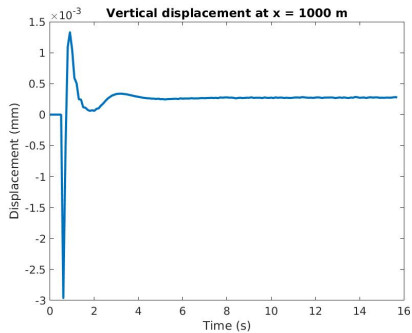


Simulation 3

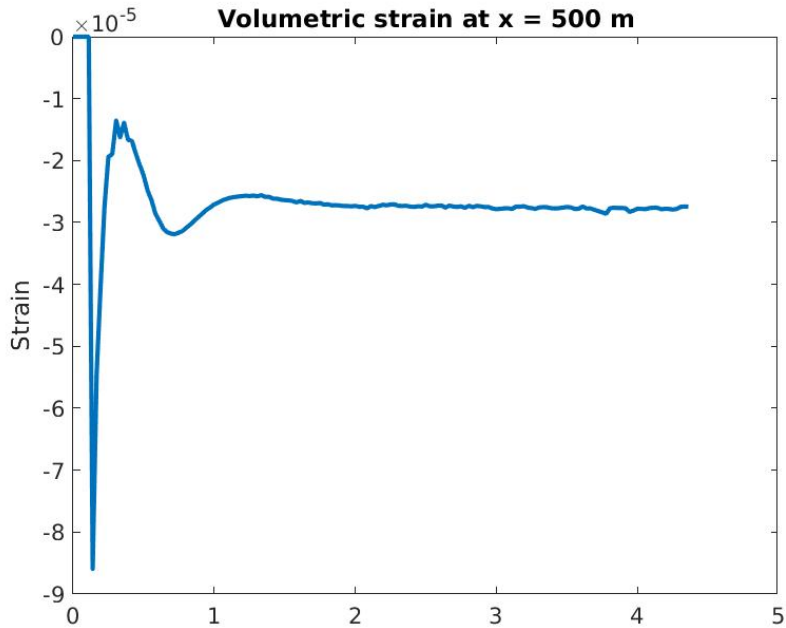


Simulation 2

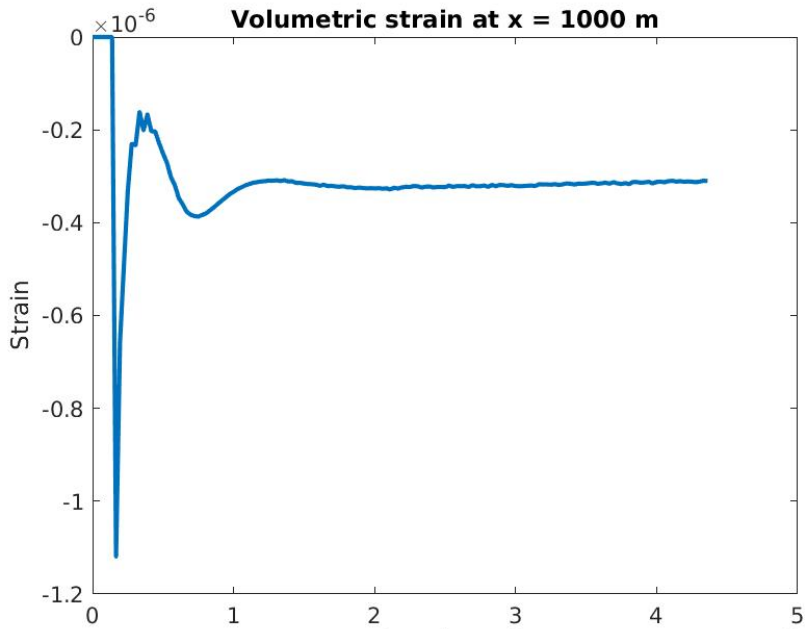
gas input, slightly higher velocity



Simulation 2



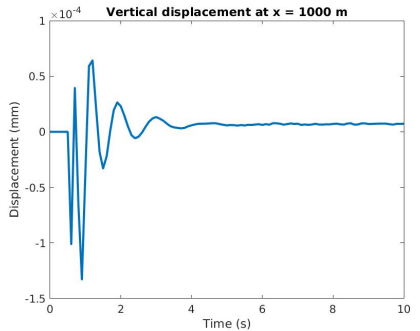
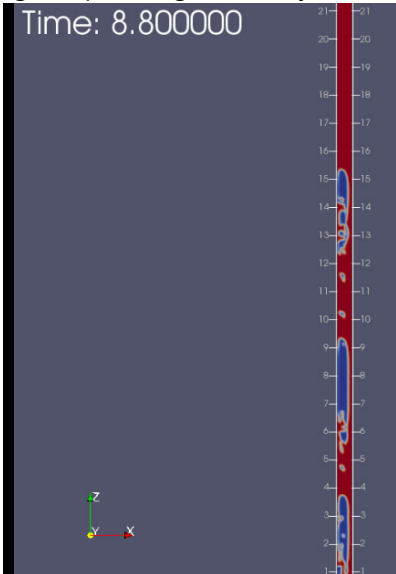
Simulation 2



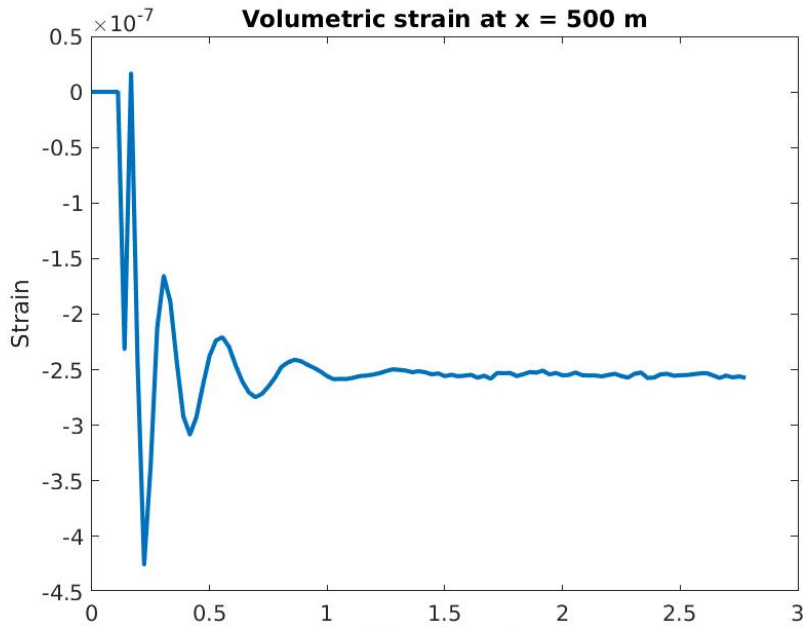
Simulation 4

gas input, larger velocity

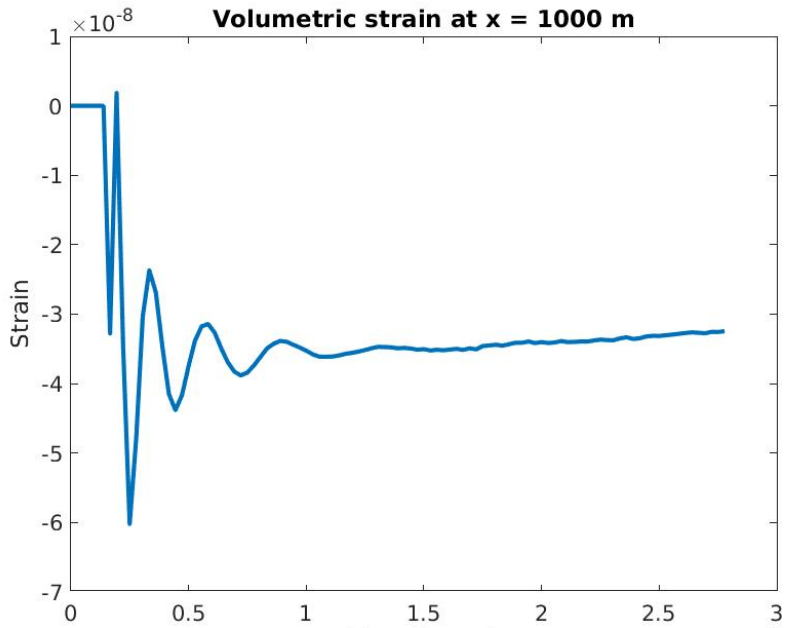
Time: 8.800000



Simulation 4



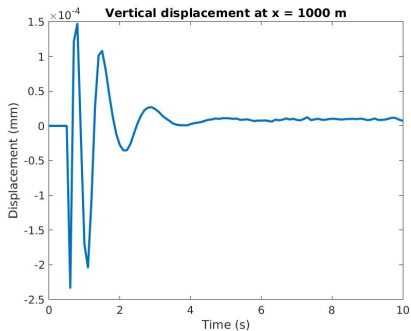
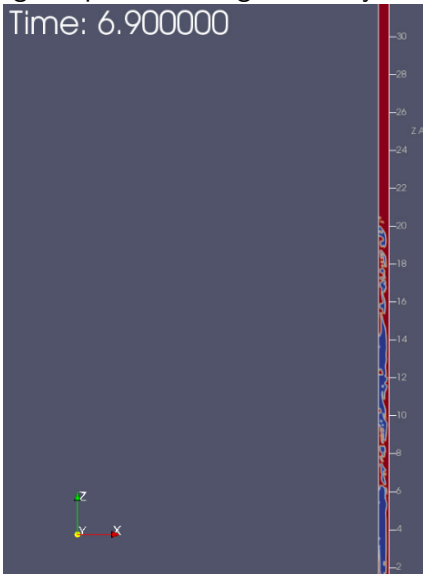
Simulation 4



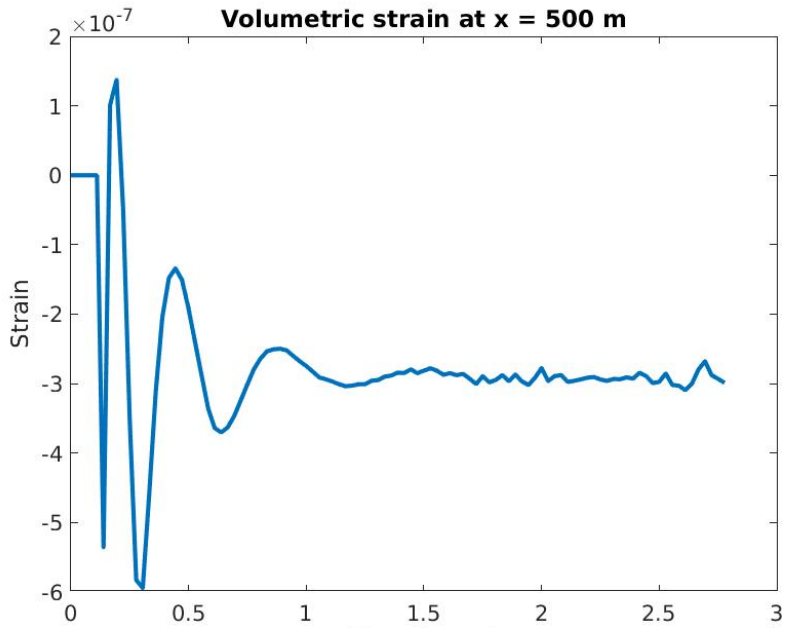
Simulation 5

gas input, even larger velocity

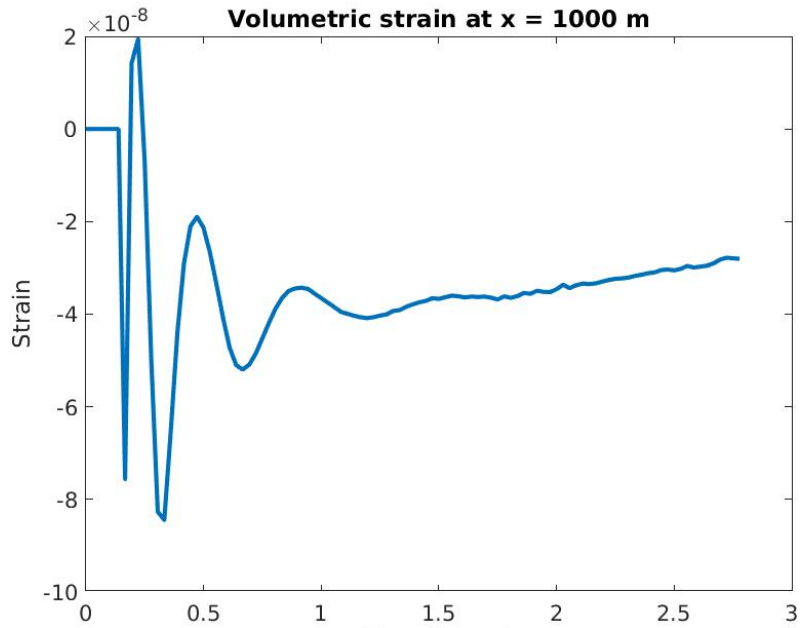
Time: 6.900000



Simulation 5



Simulation 5



- initial compression is the dominant source
- initial oscillation changes sign depending on station location
- frequencies around 1 Hz
- seems like larger input velocities result in larger amplitudes, similar waveforms (sims 4 and 5)

Next

- add (simplified) topography
- understand impact of initial transient - initial condition
- 3D
- larger diameters
- higher magma viscosities - higher gas densities